

Experiment Results

Safe Path Planning Under Perception Uncertainty

Scan Depths: 5 · 10 · 15 · 20 | Theta: 60° | Episodes: 500

1. Statistical Results

1.1 Correlation Analysis: Perception Entropy vs. Episode Reward

Spearman rank correlation (ρ) between Perception Entropy (H) and Episode Reward (R), with Bonferroni correction applied across 8 tests.

Agent	Scan Depth	Spearman (ρ)	Original p	Adjusted p (Bonferroni)
Q-Learning	5	0.6074	5.56e-52	4.45e-51
Q-Learning	10	0.5134	1.73e-35	1.38e-34
Q-Learning	15	0.4907	1.99e-32	1.59e-31
Q-Learning	20	0.6322	3.62e-57	2.89e-56
SARSA	5	0.6139	7.85e-53	6.28e-52
SARSA	10	0.7077	3.59e-76	2.87e-75
SARSA	15	0.7042	5.52e-75	4.42e-74
SARSA	20	0.7539	6.45e-93	5.16e-92

1.2 Safety Margin Saturation — Mann-Whitney U Test

Non-parametric pairwise comparison ($\alpha = 0.05$) of safety margins across consecutive scan depths.

Agent	Comparison	U-Statistic	p-value	Mean Diff	Result
Q-Learning	Scan 5 vs 10	120,238.00	0.297	+0.17	Not Significant
Q-Learning	Scan 10 vs 15	99,461.50	2.24e-08	+0.85	SIGNIFICANT
Q-Learning	Scan 15 vs 20	121,912.00	0.499	+0.23	Not Significant
SARSA	Scan 5 vs 10	86,407.50	2.89e-17	+1.33	SIGNIFICANT
SARSA	Scan 10 vs 15	128,815.50	0.403	-0.06	Not Significant
SARSA	Scan 15 vs 20	124,288.50	0.876	+0.04	Not Significant

1.3 Summary Findings

- **Correlation:** Highly significant positive correlation between perception entropy and reward across all settings ($p_{adj} < 0.001$).
- **Saturation — Q-Learning:** Safety performance saturates at Scan 15 (Scan 10 vs 15 significant, Scan 15 vs 20 not significant).
- **Saturation — SARSA:** Peak safety reached earlier at Scan 10 (Scan 5 vs 10 significant, subsequent comparisons not significant).

2. Data Previews (First 5 Rows × First 10 Episodes)

Safety Comparison — Q-Learning

Metric	Ep.1	Ep.2	Ep.3	Ep.4	Ep.5	Ep.6	Ep.7	Ep.8	Ep.9	Ep.10
Scan_5	10.19	16.12	11.25	14.38	12.08	13.42	12.49	7.16	14.52	12.95
Scan_10	18.82	14.28	7.97	17.87	10.70	14.50	10.21	8.01	19.79	12.32
Scan_15	18.40	20.08	16.59	8.15	16.15	7.88	7.00	13.42	10.88	18.32
Scan_20	10.20	19.88	17.22	18.46	20.85	21.54	19.39	19.57	5.21	8.48
Abs_Diff_5_10	8.62	1.84	3.28	3.49	1.38	1.08	2.28	0.85	5.27	0.63

First 5 rows × first 10 episode columns (7 rows × 501 columns total)

Safety Comparison — SARSA

Metric	Ep.1	Ep.2	Ep.3	Ep.4	Ep.5	Ep.6	Ep.7	Ep.8	Ep.9	Ep.10
Scan_5	11.49	13.67	9.44	8.24	9.83	12.18	6.98	10.62	7.38	14.66
Scan_10	16.21	19.91	16.75	18.50	13.48	8.33	13.41	11.06	9.94	16.07
Scan_15	18.06	10.43	7.31	13.77	16.61	10.56	21.41	12.46	9.96	9.82
Scan_20	13.66	21.62	16.90	14.03	19.99	22.69	15.51	10.48	16.68	10.45
Abs_Diff_5_10	4.73	6.24	7.30	10.26	3.65	3.85	6.43	0.44	2.56	1.40

First 5 rows × first 10 episode columns (7 rows × 501 columns total)

Q-Learning History — Scan 5

Metric	Ep.376	Ep.377	Ep.378	Ep.379	Ep.380	Ep.381	Ep.382	Ep.383	Ep.384	Ep.385	Ep.386
Reward	37.32	39.19	8.95	13.98	55.05	1231.34	1179.11	1213.36	1220.04	1285.93	1213.39
Path_Length	400	400	400	400	400	240	294	263	258	194	257
TD_Error	2.68	2.38	2.60	2.51	2.59	6.53	6.01	7.66	7.26	7.55	6.36
Goal_Reached	0	0	0	0	0	1	1	1	1	1	1
Avg_Safety_Mar...	11.13	11.30	11.17	10.72	10.78	12.48	11.19	11.71	12.44	12.80	11.13

Convergence at Episode_381 (ep 381) — showing ±5 episodes around convergence (7 metrics × 500 episodes total)

Q-Learning History — Scan 10

Metric	Ep.389	Ep.390	Ep.391	Ep.392	Ep.393	Ep.394	Ep.395	Ep.396	Ep.397	Ep.398	Ep.399
Reward	22.42	-1.68	11.17	46.20	-82.45	1088.96	46.53	1149.75	-103.11	53.70	1196.81
Path_Length	400	400	400	400	400	372	400	320	400	400	275
TD_Error	2.46	2.46	2.64	2.52	2.88	5.15	2.46	5.52	2.82	2.50	6.20
Goal_Reached	0	0	0	0	0	1	0	1	0	0	1
Avg_Safety_Mar...	12.76	12.01	12.21	12.52	11.48	12.64	13.06	12.89	10.87	12.14	11.73

Convergence at Episode_394 (ep 394) — showing ±5 episodes around convergence (7 metrics × 500 episodes total)

Q-Learning History — Scan 15

Metric	Ep.361	Ep.362	Ep.363	Ep.364	Ep.365	Ep.366	Ep.367	Ep.368	Ep.369	Ep.370	Ep.371
Reward	24.37	42.21	31.98	-59.31	19.24	1073.65	1217.09	1173.78	1262.35	1225.43	1091.33
Path_Length	400	400	400	400	400	395	264	305	220	252	375
TD_Error	3.20	2.72	2.86	2.75	2.76	5.04	6.61	6.08	7.21	6.89	5.05

Metric	Ep.361	Ep.362	Ep.363	Ep.364	Ep.365	Ep.366	Ep.367	Ep.368	Ep.369	Ep.370	Ep.371
Goal_Reached	0	0	0	0	0	1	1	1	1	1	1
Avg_Safety_Mar...	12.43	12.81	13.41	12.19	13.38	13.46	12.04	12.87	13.32	12.08	13.13

Convergence at Episode_366 (ep 366) — showing ±5 episodes around convergence (7 metrics × 500 episodes total)

Q-Learning History — Scan 20

Metric	Ep.369	Ep.370	Ep.371	Ep.372	Ep.373	Ep.374	Ep.375	Ep.376	Ep.377	Ep.378	Ep.379
Reward	-1.74	31.36	16.78	-16.22	14.76	1101.42	1086.51	1232.93	1135.02	28.18	1222.84
Path_Length	400	400	400	400	400	364	375	248	330	400	257
TD_Error	2.60	2.78	2.86	2.65	2.52	5.53	5.33	6.57	5.84	2.90	6.30
Goal_Reached	0	0	0	0	0	1	1	1	1	0	1
Avg_Safety_Mar...	11.77	12.07	12.56	13.52	12.29	12.70	13.03	12.38	12.83	13.51	13.01

Convergence at Episode_374 (ep 374) — showing ±5 episodes around convergence (7 metrics × 500 episodes total)

SARSA History — Scan 5

Metric	Ep.375	Ep.376	Ep.377	Ep.378	Ep.379	Ep.380	Ep.381	Ep.382	Ep.383	Ep.384	Ep.385
Reward	61.53	55.32	34.32	-23.14	24.59	1086.20	1183.94	1265.89	1131.53	1184.64	1261.96
Path_Length	400	400	400	400	400	390	285	216	354	297	212
TD_Error	2.81	2.80	3.03	2.95	2.78	5.50	6.55	7.53	5.84	6.30	7.42
Goal_Reached	0	0	0	0	0	1	1	1	1	1	1
Avg_Safety_Mar...	10.94	10.18	9.84	9.58	10.58	10.80	10.77	11.76	11.74	11.04	11.79

Convergence at Episode_380 (ep 380) — showing ±5 episodes around convergence (7 metrics × 500 episodes total)

SARSA History — Scan 10

Metric	Ep.350	Ep.351	Ep.352	Ep.353	Ep.354	Ep.355	Ep.356	Ep.357	Ep.358	Ep.359	Ep.360
Reward	-113.97	13.67	60.56	55.85	-107.81	1143.99	44.06	1119.85	10.70	1237.42	1189.90
Path_Length	400	400	400	400	400	329	400	344	400	245	283
TD_Error	2.67	2.85	2.80	2.85	2.90	5.84	2.82	5.84	2.77	6.84	6.41
Goal_Reached	0	0	0	0	0	1	0	1	0	1	1
Avg_Safety_Mar...	10.56	11.89	12.90	12.71	10.62	13.01	11.96	12.11	12.64	13.80	12.28

Convergence at Episode_355 (ep 355) — showing ±5 episodes around convergence (7 metrics × 500 episodes total)

SARSA History — Scan 15

Metric	Ep.367	Ep.368	Ep.369	Ep.370	Ep.371	Ep.372	Ep.373	Ep.374	Ep.375	Ep.376	Ep.377
Reward	32.99	48.31	31.25	28.92	63.06	1079.06	48.34	1170.56	1230.21	1188.13	1250.09
Path_Length	400	400	400	400	400	393	400	301	249	291	223
TD_Error	2.97	3.08	2.81	2.79	2.88	5.33	2.93	6.34	6.99	6.01	7.11
Goal_Reached	0	0	0	0	0	1	0	1	1	1	1
Avg_Safety_Mar...	12.86	13.51	12.34	12.81	14.17	13.33	12.89	13.14	12.67	13.53	12.12

Convergence at Episode_372 (ep 372) — showing ±5 episodes around convergence (7 metrics × 500 episodes total)

SARSA History — Scan 20

Metric	Ep.383	Ep.384	Ep.385	Ep.386	Ep.387	Ep.388	Ep.389	Ep.390	Ep.391	Ep.392	Ep.393
Reward	47.69	34.15	8.29	32.58	44.35	1253.27	-25.74	1217.83	1236.32	1187.36	1274.72
Path_Length	400	400	400	400	400	230	400	262	250	288	216
TD_Error	2.83	2.98	2.79	2.91	2.91	7.31	2.98	6.72	6.47	5.97	6.78
Goal_Reached	0	0	0	0	0	1	0	1	1	1	1
Avg_Safety_Mar...	13.11	13.83	14.06	13.04	12.34	13.95	12.37	13.58	13.77	13.60	14.48

Convergence at Episode_388 (ep 388) — showing ±5 episodes around convergence (7 metrics × 500 episodes total)

3. Figures

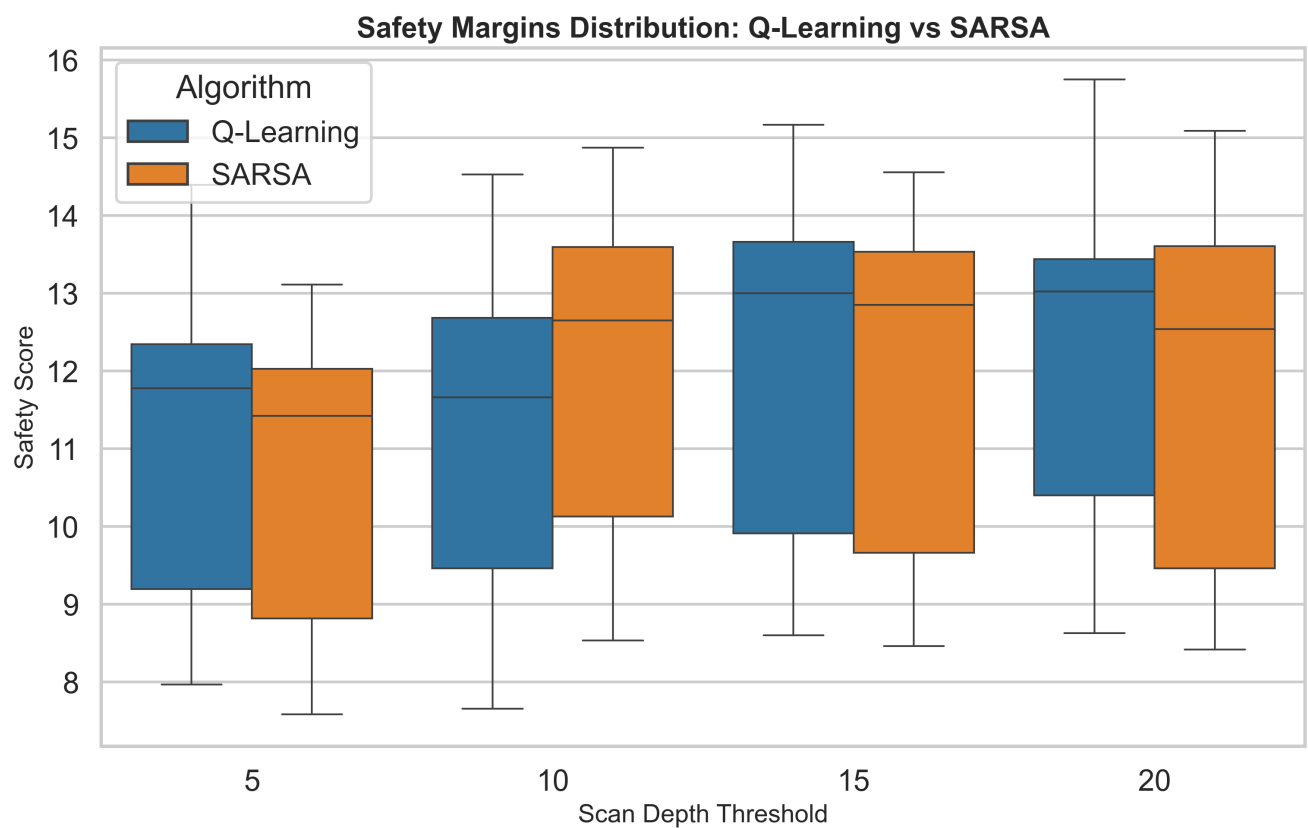


Figure: Box Plot — Safety Margins per Scan Depth (Q-Learning vs SARSA)

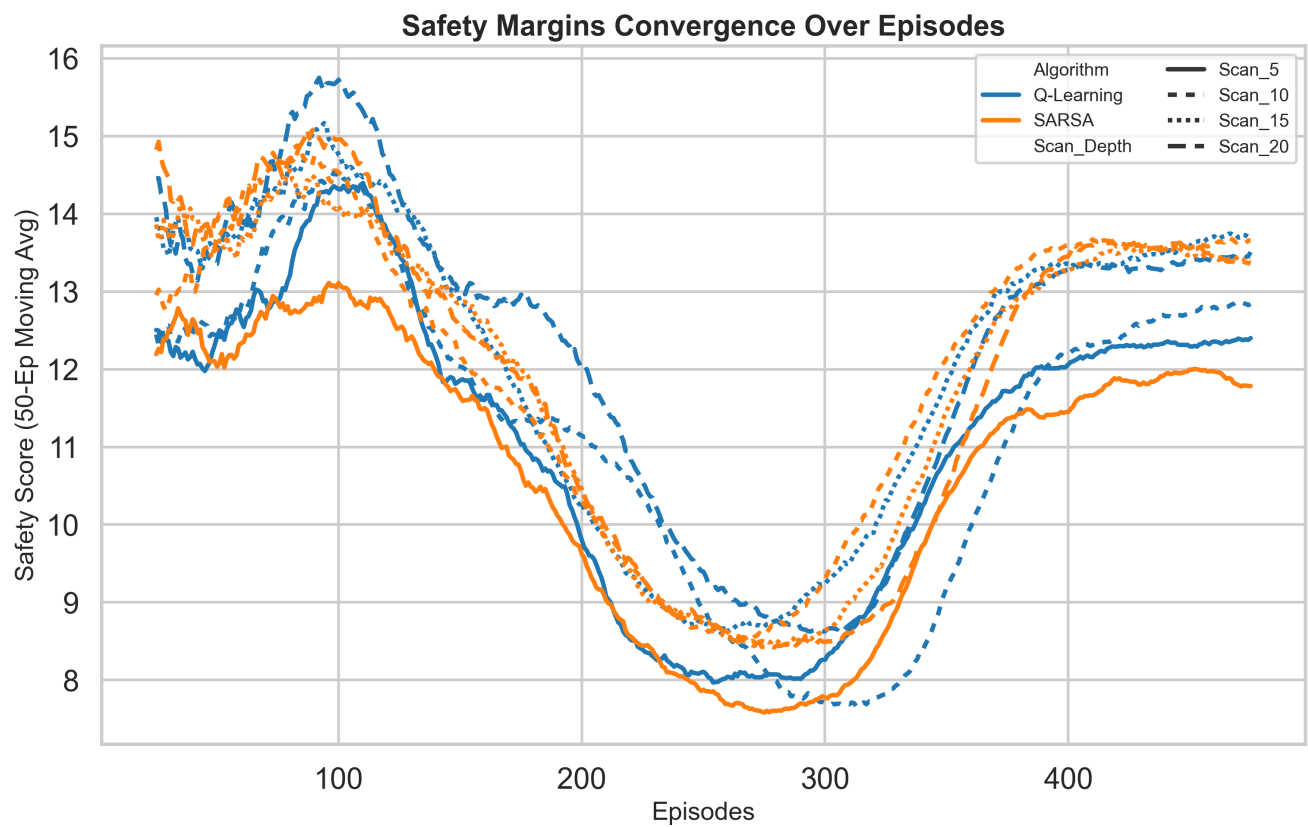


Figure: Safety Margins Overview

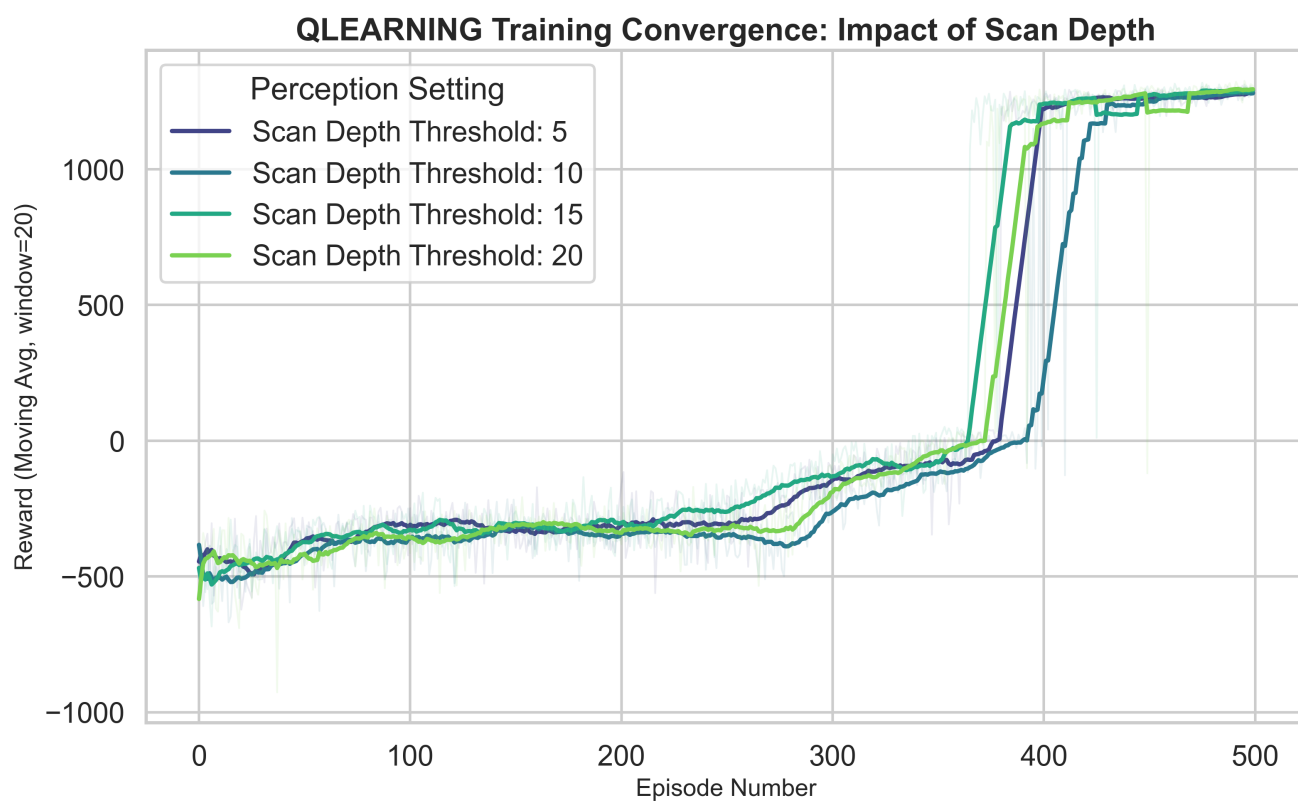


Figure: Fig 1.1 — Q-Learning: Reward Convergence across Scan Depths

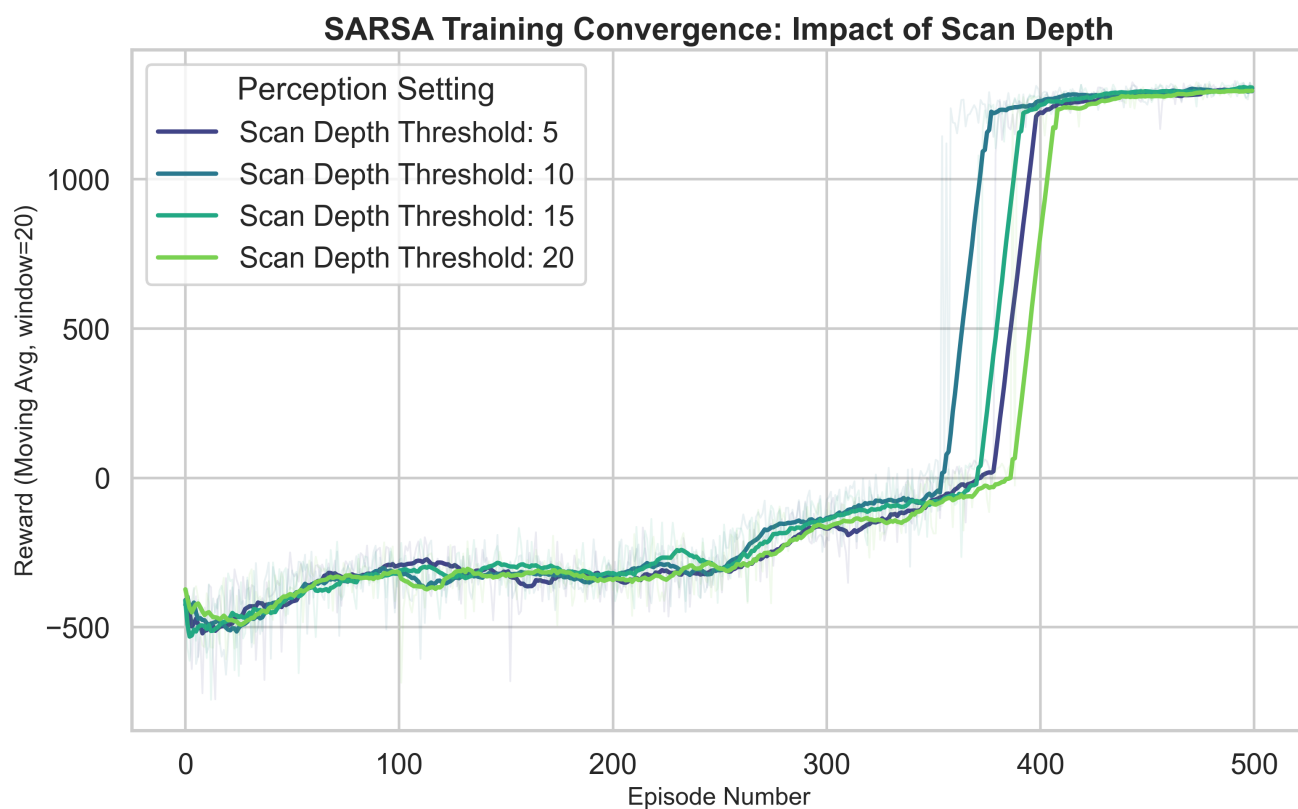


Figure: Fig 1.1 — SARSA: Reward Convergence across Scan Depths

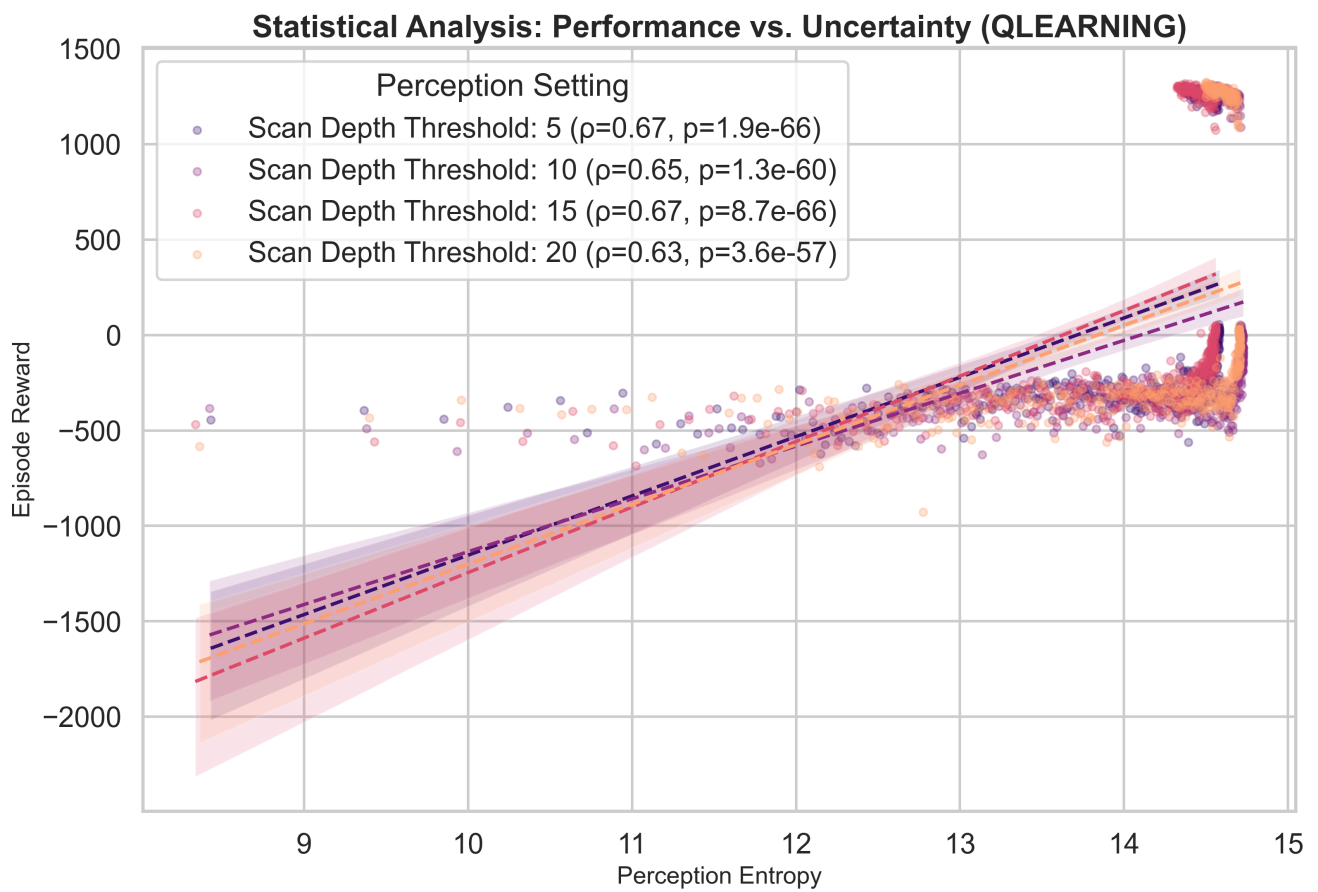


Figure: Fig 1.2 — Q-Learning: Reward vs Perception Entropy

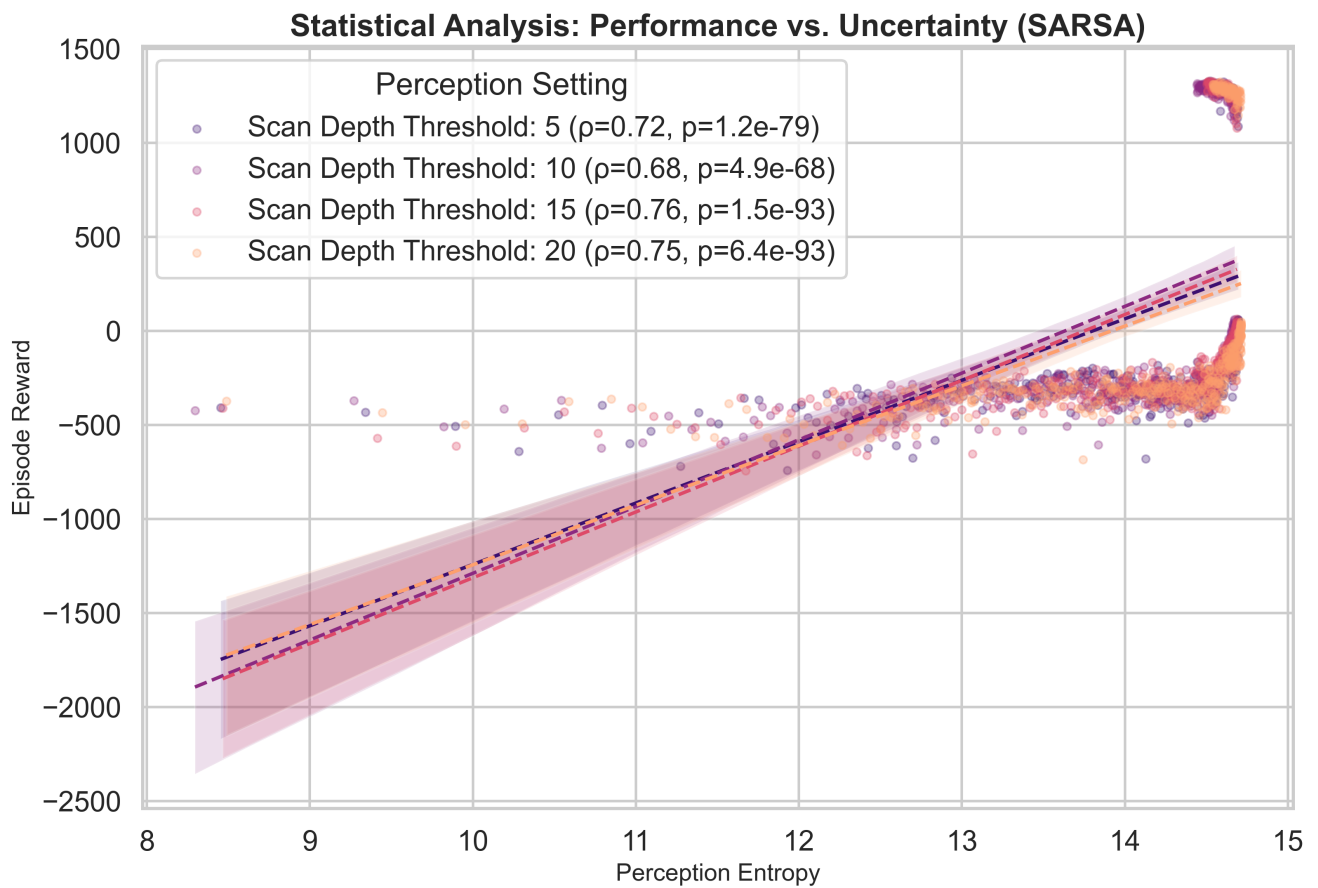


Figure: Fig 1.2 — SARSA: Reward vs Perception Entropy

Algorithm Comparison (Theta: 60, Scan Depth: 5)

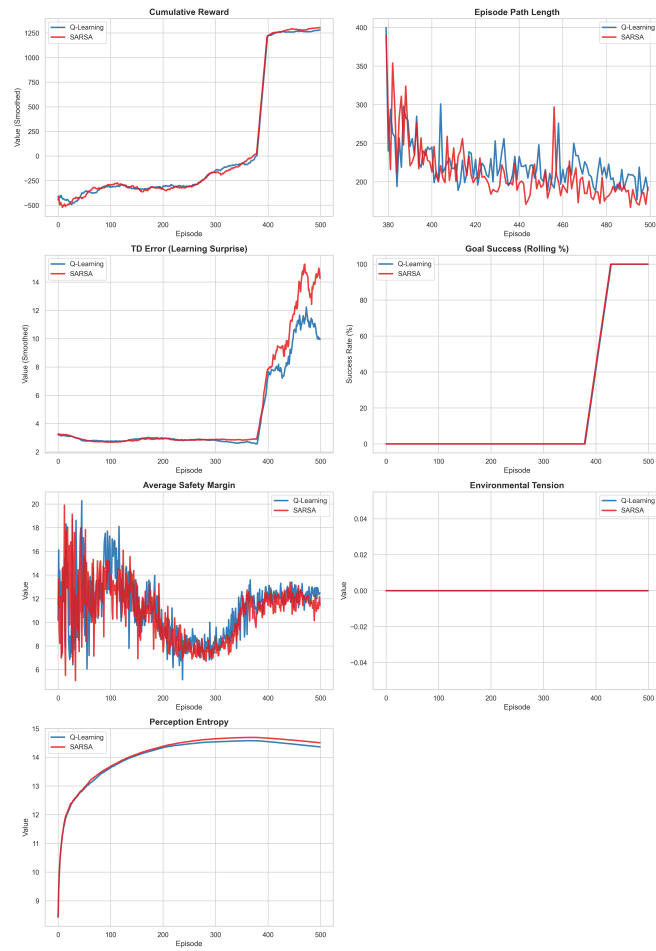


Figure: Algorithm Comparison — Theta 60, Scan Depth 5

Algorithm Comparison (Theta: 60, Scan Depth: 10)

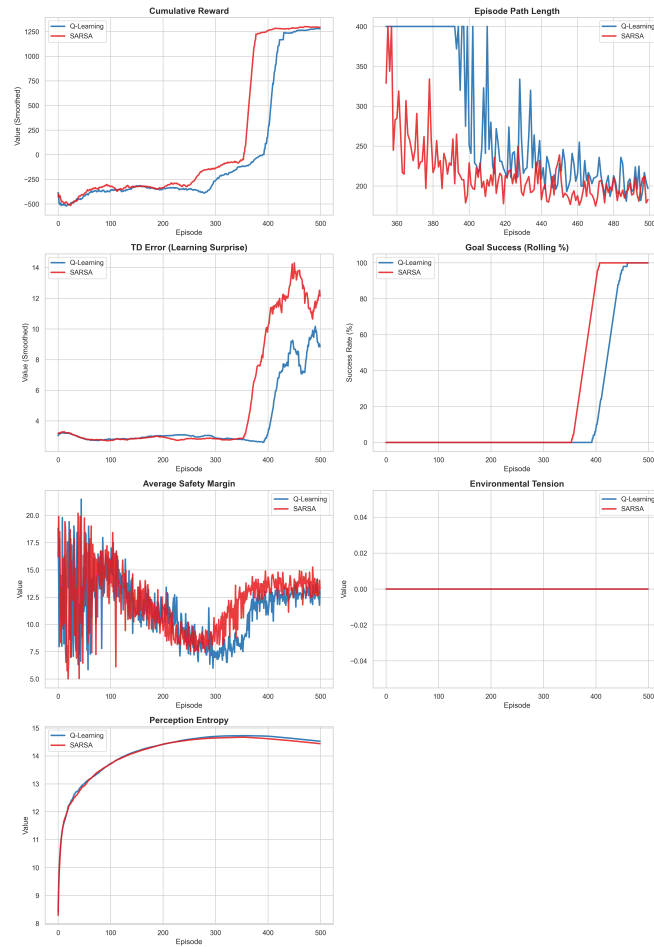


Figure: Algorithm Comparison — Theta 60, Scan Depth 10

Algorithm Comparison (Theta: 60, Scan Depth: 15)

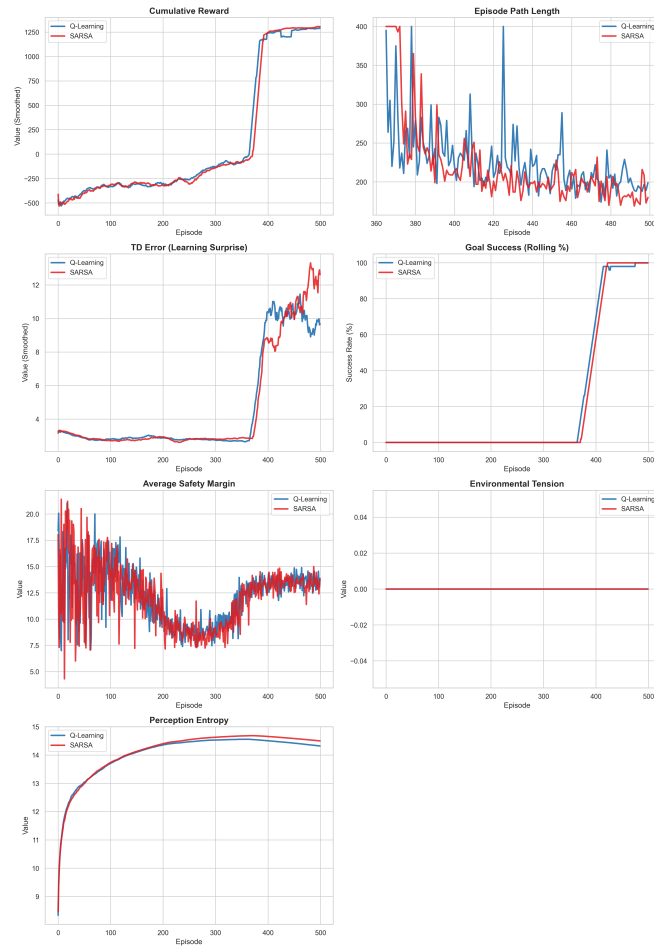


Figure: Algorithm Comparison — Theta 60, Scan Depth 15

Algorithm Comparison (Theta: 60, Scan Depth: 20)

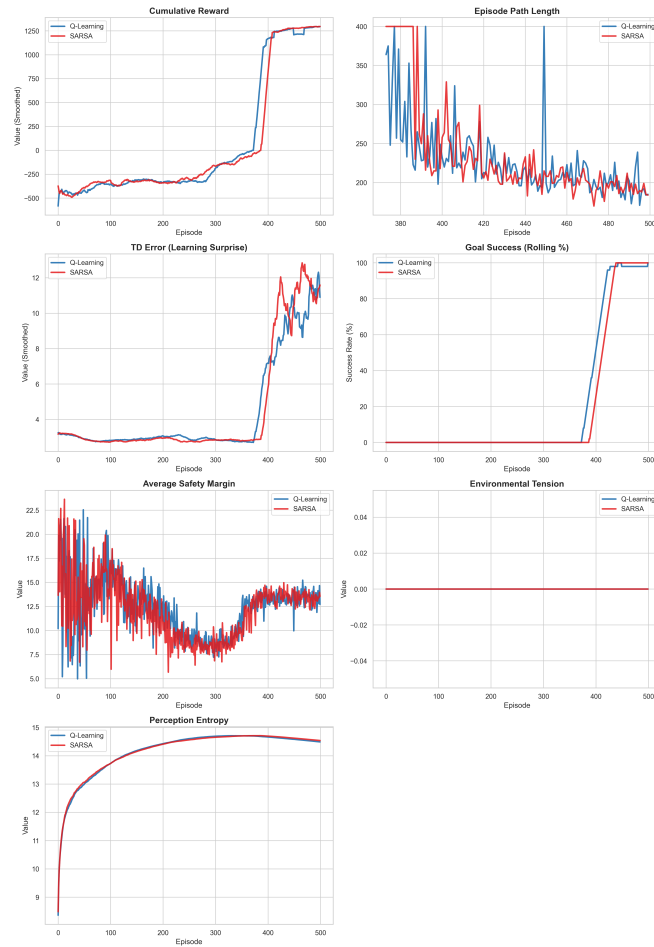


Figure: Algorithm Comparison — Theta 60, Scan Depth 20

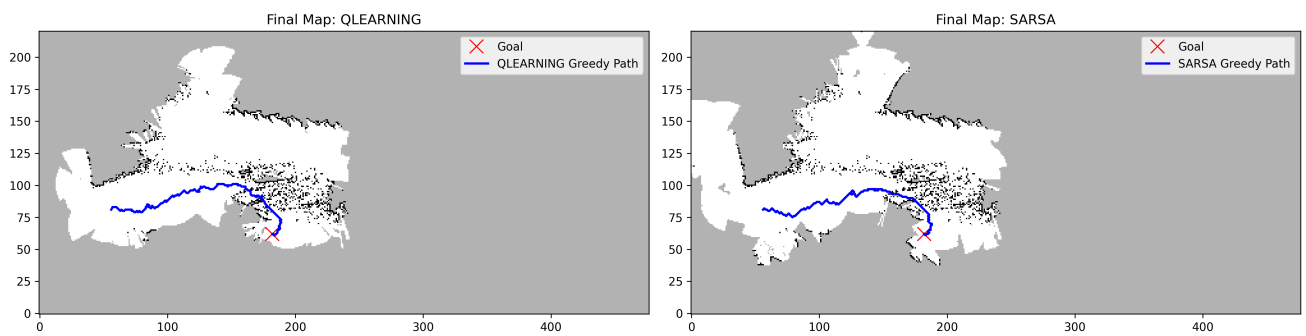


Figure: RL Mode 4 Comparison — Theta 60, Scan Depth 5

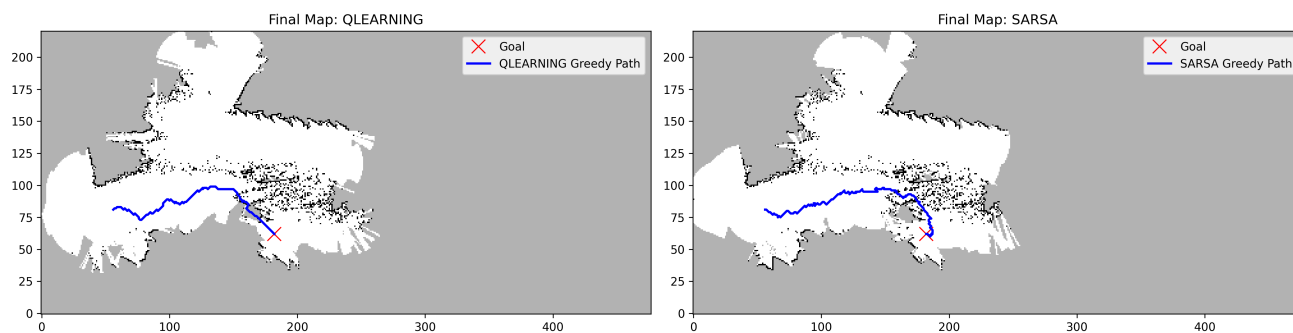


Figure: RL Mode 4 Comparison — Theta 60, Scan Depth 10

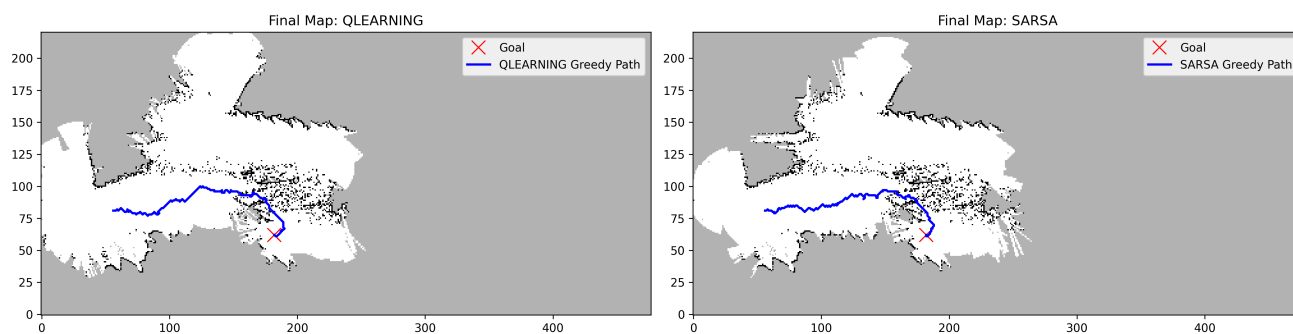


Figure: RL Mode 4 Comparison — Theta 60, Scan Depth 15

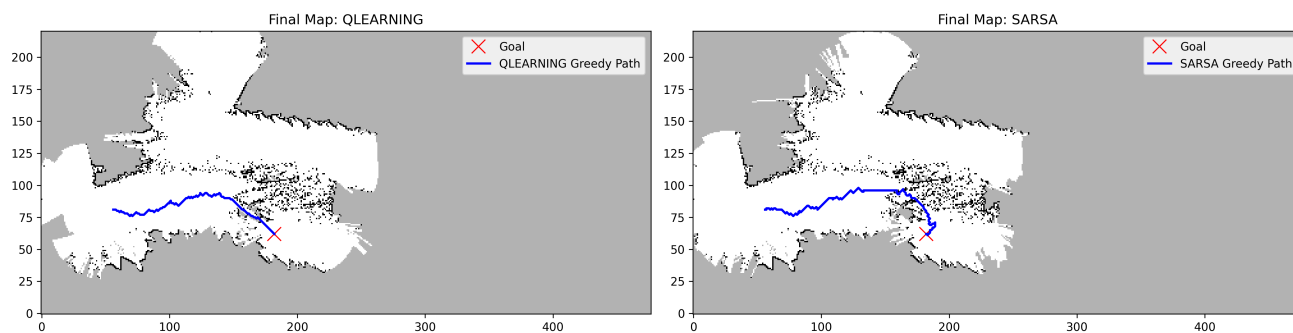


Figure: RL Mode 4 Comparison — Theta 60, Scan Depth 20

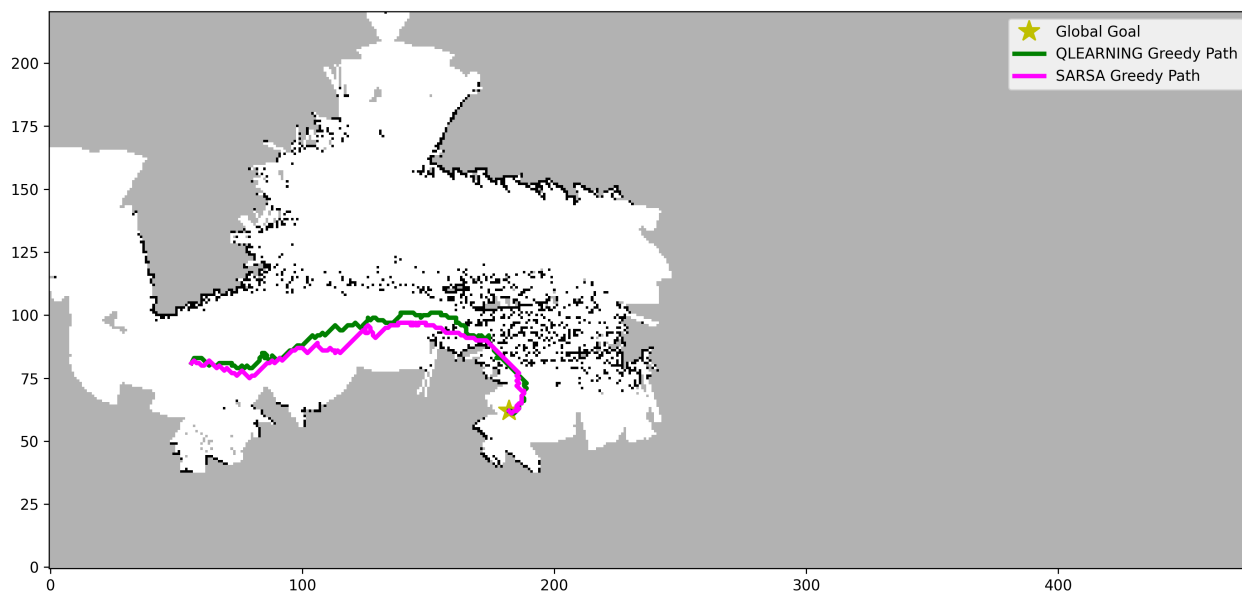


Figure: Path Overlay — Theta 60, Scan Depth 5

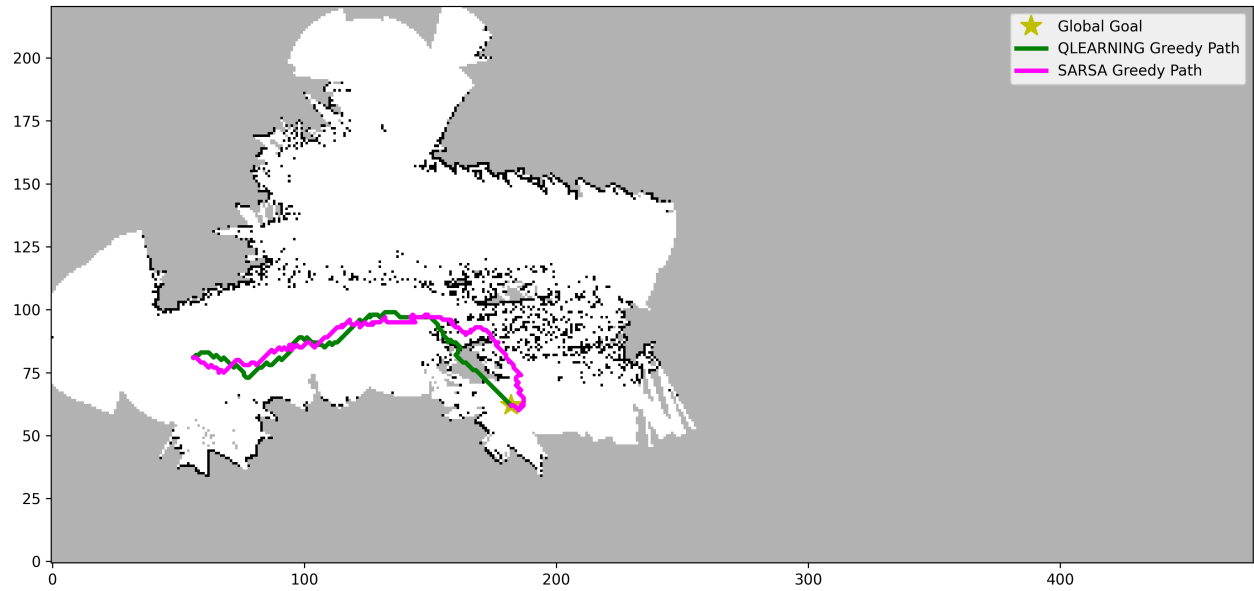


Figure: Path Overlay — Theta 60, Scan Depth 10

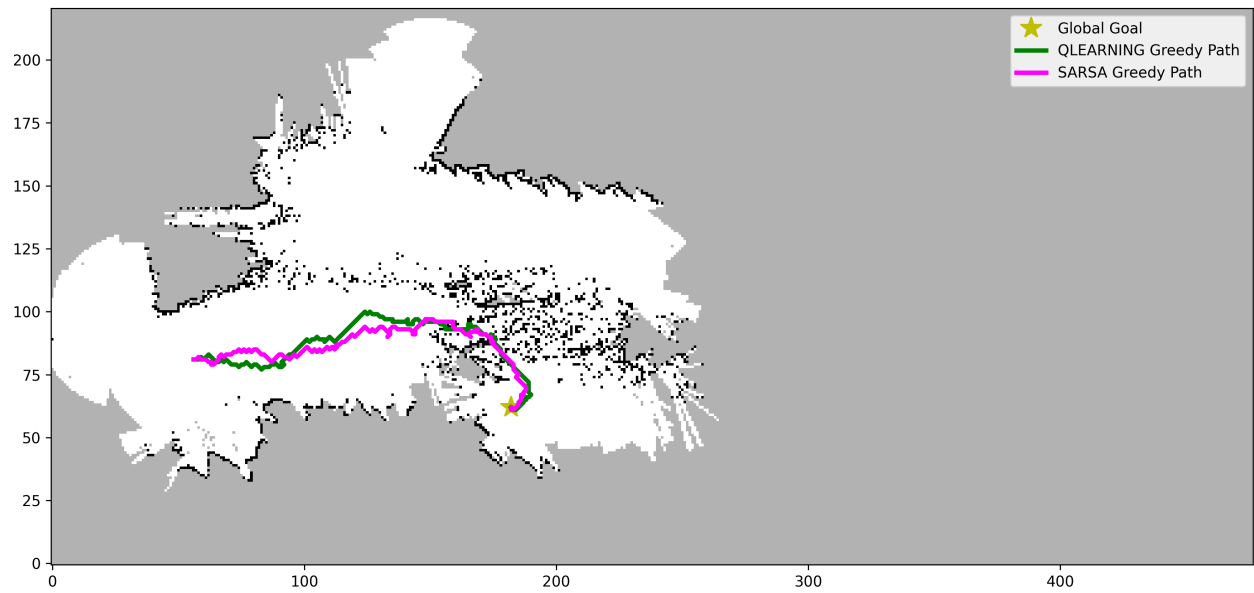


Figure: Path Overlay — Theta 60, Scan Depth 15

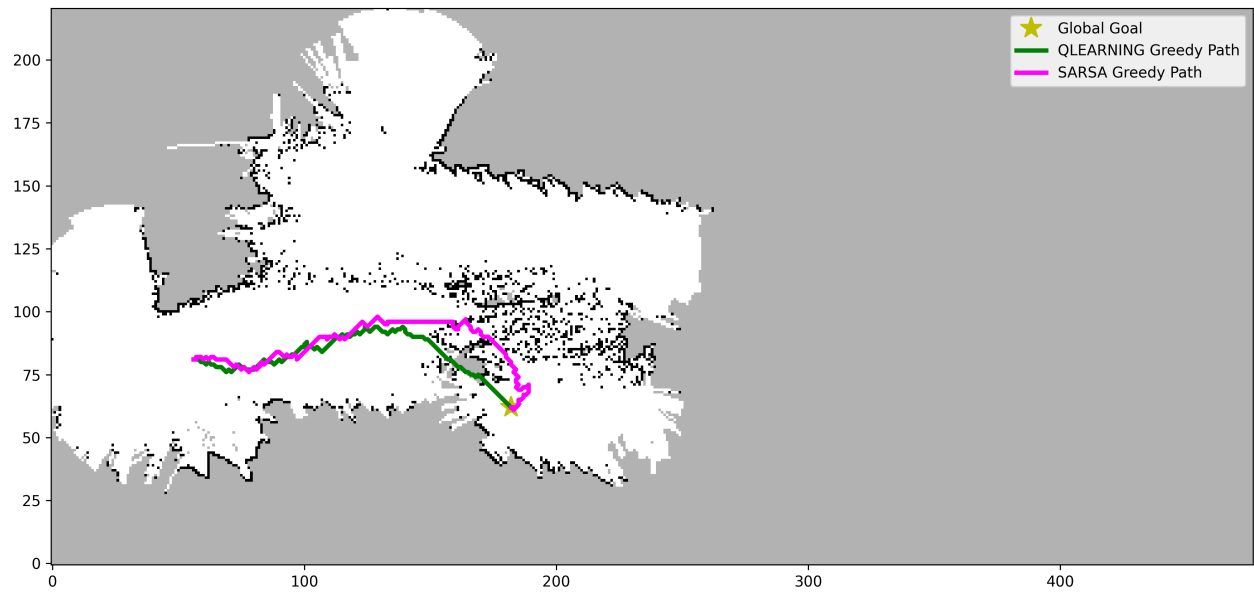


Figure: Path Overlay — Theta 60, Scan Depth 20